# FR3068E-C 开发板使用手册

FreqChip Confi

版本号:V1.0
发布日期:2024.7.31

FREQCHIP
富芮坤

# 目录

1. FR3068E-C 开发板硬件设计 ..... 3
1.1 FR3068E-C 开发板系统框图 ..... 3
1.2 开发板硬件资源 ..... 4
1.3 调试口说明 ..... 4
1.4 LCD 屏接口说明 ..... 4
2. FR3068E-C 开发板特性 ..... 6
3. 开发板使用说明 ..... 7
3.1 FR3068E-C 开发板连接调试器数据线、电源线 ..... 7
3.2 串口板接上电脑，打开 PC 端烧录工具 ..... 8
联系信息 ..... 11
版本修订 ..... 11

FreqChip Confidential

# 1. FR3068E-C 开发板硬件设计

## 1.1 FR3068E-C 开发板系统框图

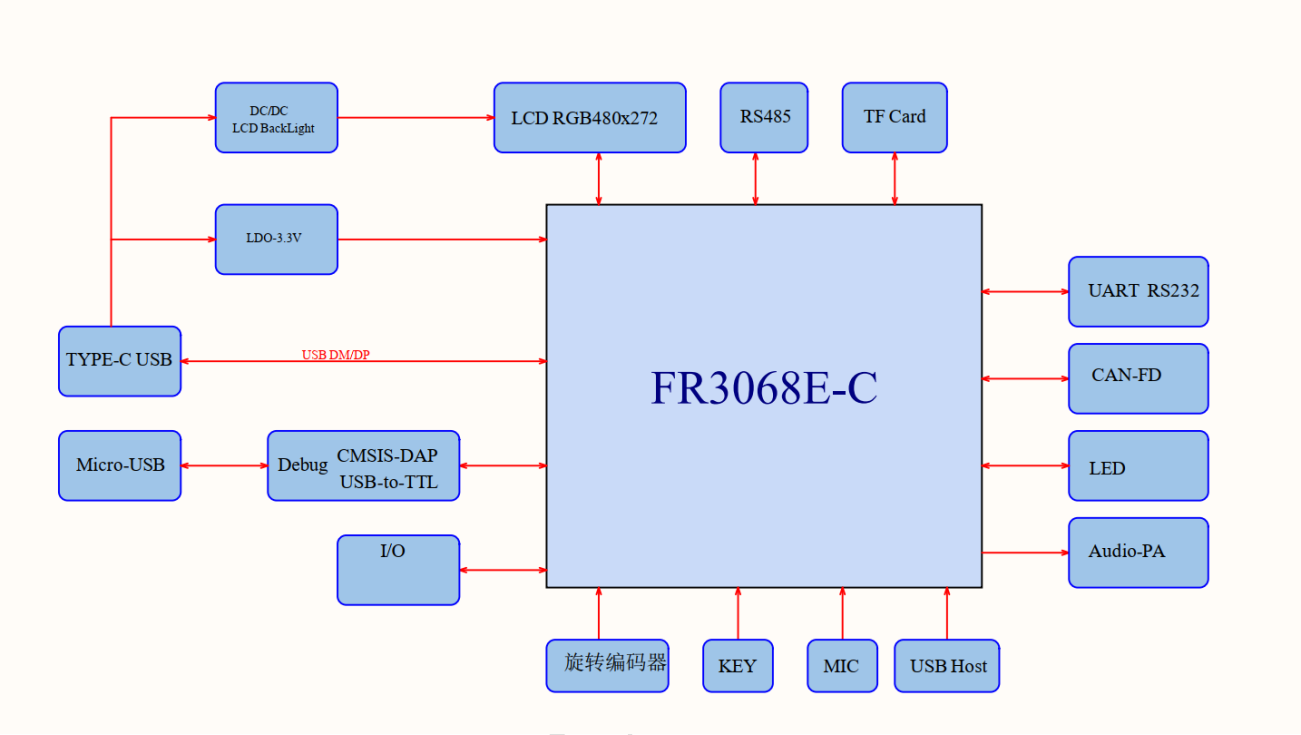



图 1-1 FR3068E 开发板系统框图

FreqChip

## 1.2 开发板硬件资源

开发板硬件资源分配如图 1-2 所示：

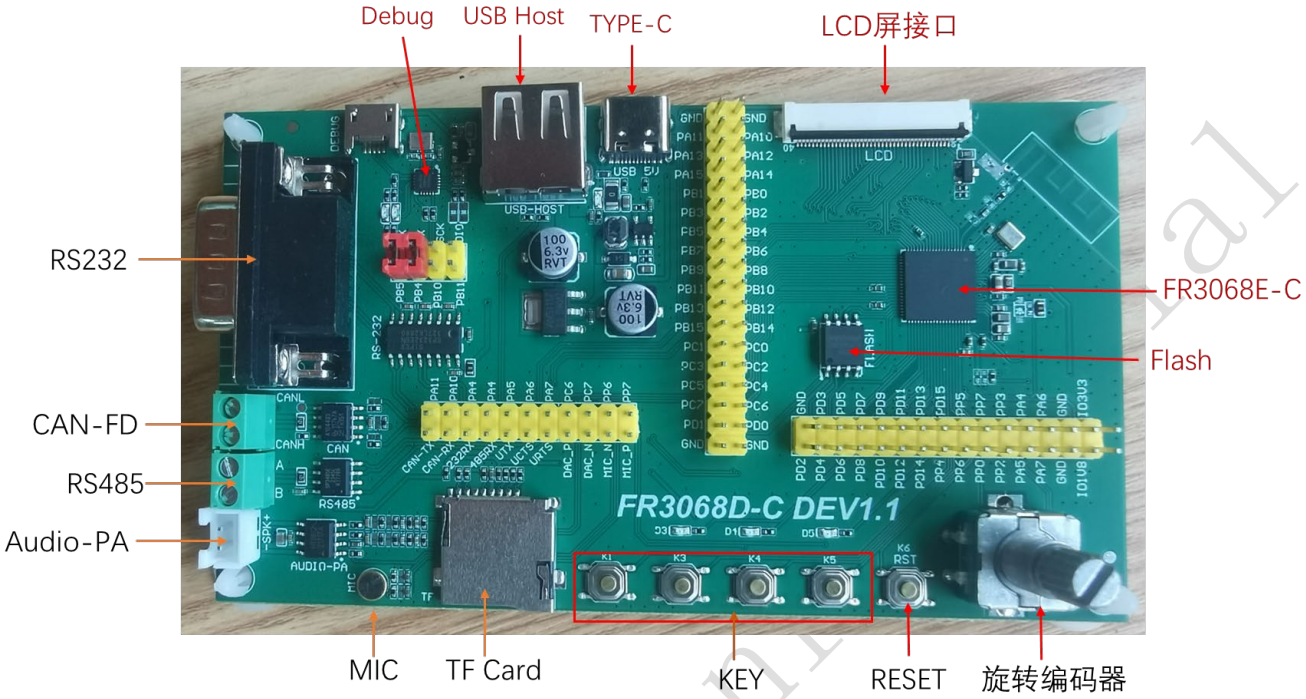



图 1-2 开发板硬件资源分配图

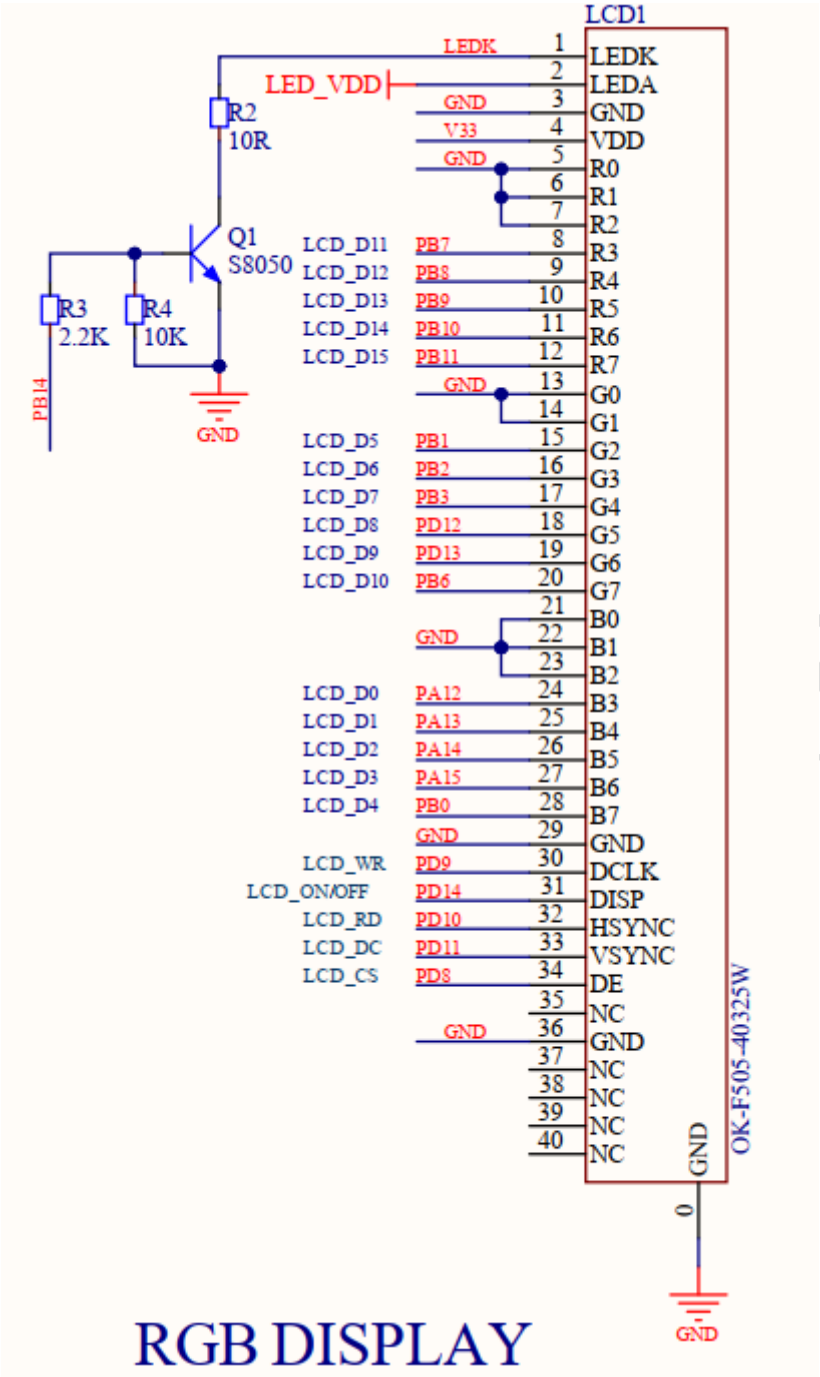
## 1.3 调试口说明

- RX：调试器 RX
- TX：调试器 TX
- SCK：调试器 CMSIS-DAP CLK
- DIO：调试器 CMSIS-DAP DIO
- PB5：主控 TX
- PB4：主控 TX
- PB10：主控 SWD CLK
- PB11：主控 SWD DIO

## 1.4 LCD 屏接口说明

RGB 屏接口定义如下：

FreqChip Confidential


LCD1
LEDK
LED_VDD
R2
10R
Q1
S8050
R3
2.2K
R4
10K
PB14
GND
V33
LCD_D11
LCD_D12
LCD_D13
LCD_D14
LCD_D15
LCD_D5
LCD_D6
LCD_D7
LCD_D8
LCD_D9
LCD_D10
LCD_D0
LCD_D1
LCD_D2
LCD_D3
LCD_D4
LCD_WR
LCD_ON/OFF
LCD_RD
LCD_DC
LCD_CS
PB7
PB8
PB9
PB10
PB11
PB1
PB2
PB3
PD12
PD13
PB6
PA12
PA13
PA14
PA15
PB0
PD9
PD14
PD10
PD11
PD8
1 LEDK
2 LEDA
3 GND
4 VDD
5 R0
6 R1
7 R2
8 R3
9 R4
10 R5
11 R6
12 R7
13 G0
14 G1
15 G2
16 G3
17 G4
18 G5
19 G6
20 G7
21 B0
22 B1
23 B2
24 B3
25 B4
26 B5
27 B6
28 B7
29 GND
30 DCLK
31 DISP
32 HSYNC
33 VSYNC
34 DE
35 NC
36 GND
37 NC
38 NC
39 NC
40 NC
OK-F505-40325W
0
RGB DISPLAY

Fre dential

# 2. FR3068E-C 开发板特性

- 蓝牙 V5.3 协议
- 串口(PB4-RX、PB5-TX)、 J-Link 调试接口(PB10-SWCLK、PB11-SWDIO)，支持 J-Link V9
- PCB 天线
- 5 个用户按键、1 个复位按键，1 个旋转编码器
- 芯片支持 2x16 路 PWM 输出
- 芯片支持 9 路 12 位 ADC
- 芯片支持 3 路 I2C
- 芯片支持 5 路 UART、支持流控功能
- 芯片支持 2 路 SPI 主接口，支持 QSPI， OSPI；2 路 SPI 做从
- 芯片支持 2 路 I2S
- 芯片支持 8080-16bit、RGB、QSPI 显示屏通信
- 芯片支持 1 路 SD 卡，支持 SDIO3.0/eMMC 4.5.1
- 芯片支持 USB OTG
- 芯片支持 4 路数字 MIC
- 芯片支持 1 路 SPDIF 输入、1 路 SPDIF 输出
- 支持 CAN-FD
- 支持 RS485
- 支持 RS232
- 芯片支持 AES-128, AES-192, AES-256

FreqChip Confidential

# 3. 开发板使用说明

## 3.1 FR3068E-C 开发板连接调试器数据线、电源线

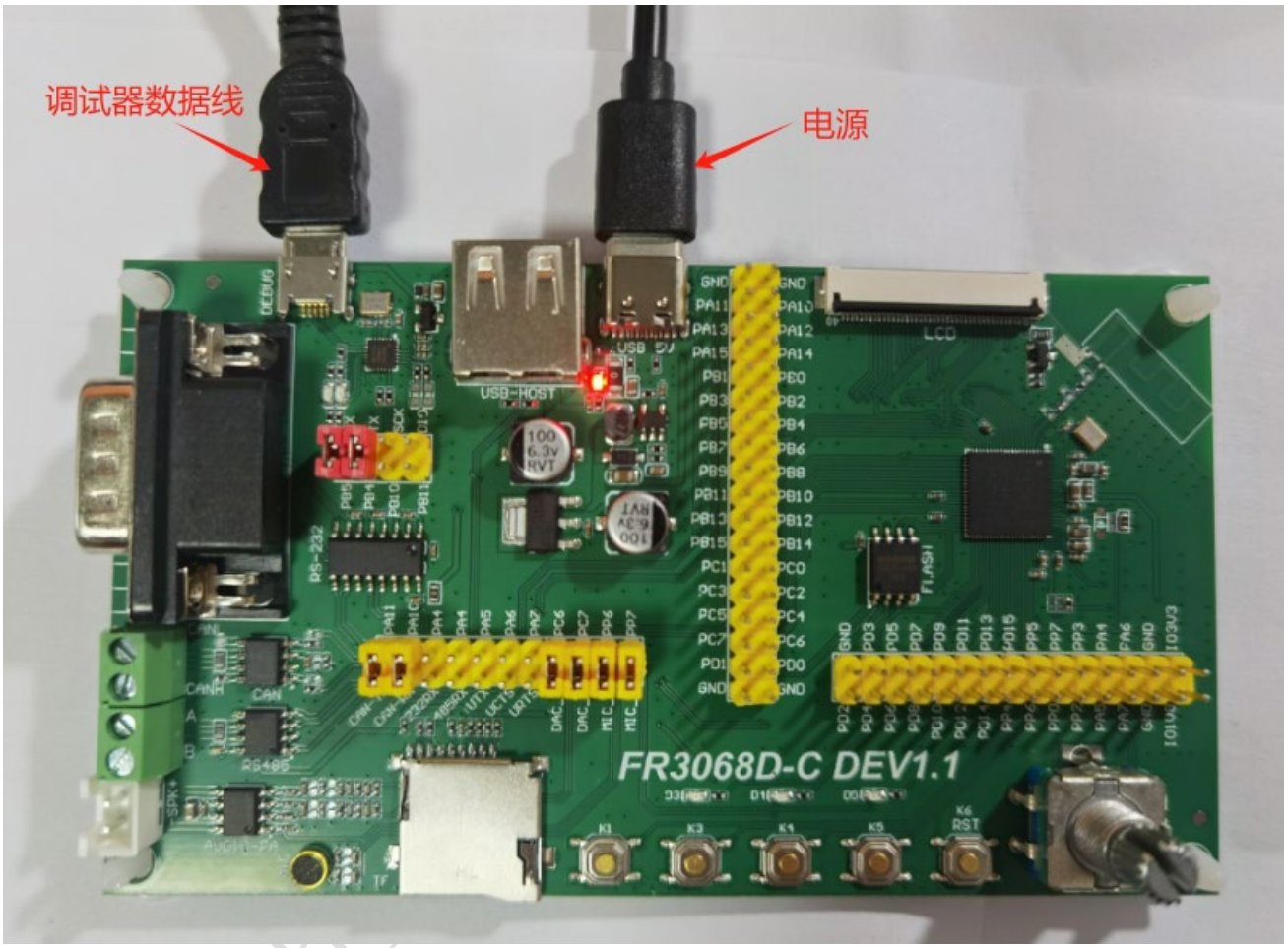



图 3-1 开发板连接示意图

FreqChip

## 3.2 串口板接上电脑，打开 PC 端烧录工具




图 3-2 烧录工具串口配置




图 3-3 烧录工具打开烧录

3.3 按开发板上面的复位键，开始烧录软件，烧录完成后并运行软件。

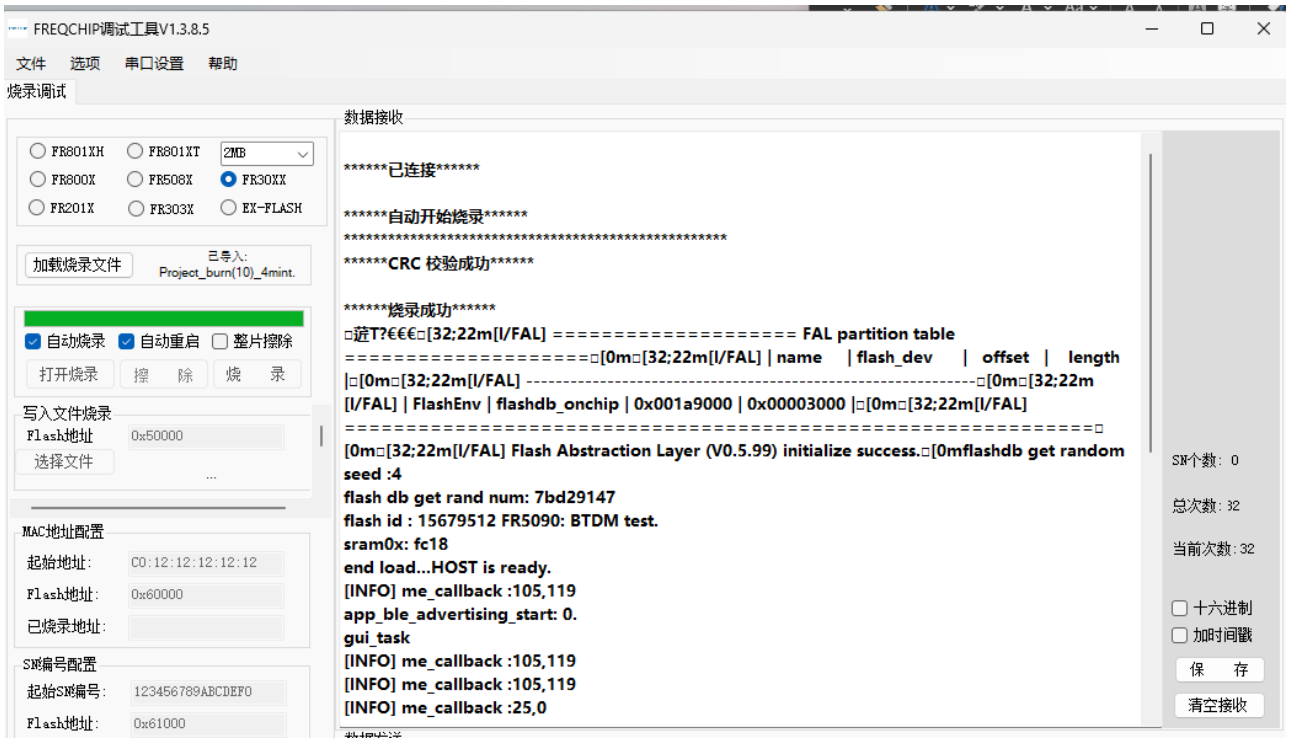



图 3-4 软件烧录成功

4、开发板原理图

见附件。

开发板原理图

开发板位号图

FreqChip Confidential

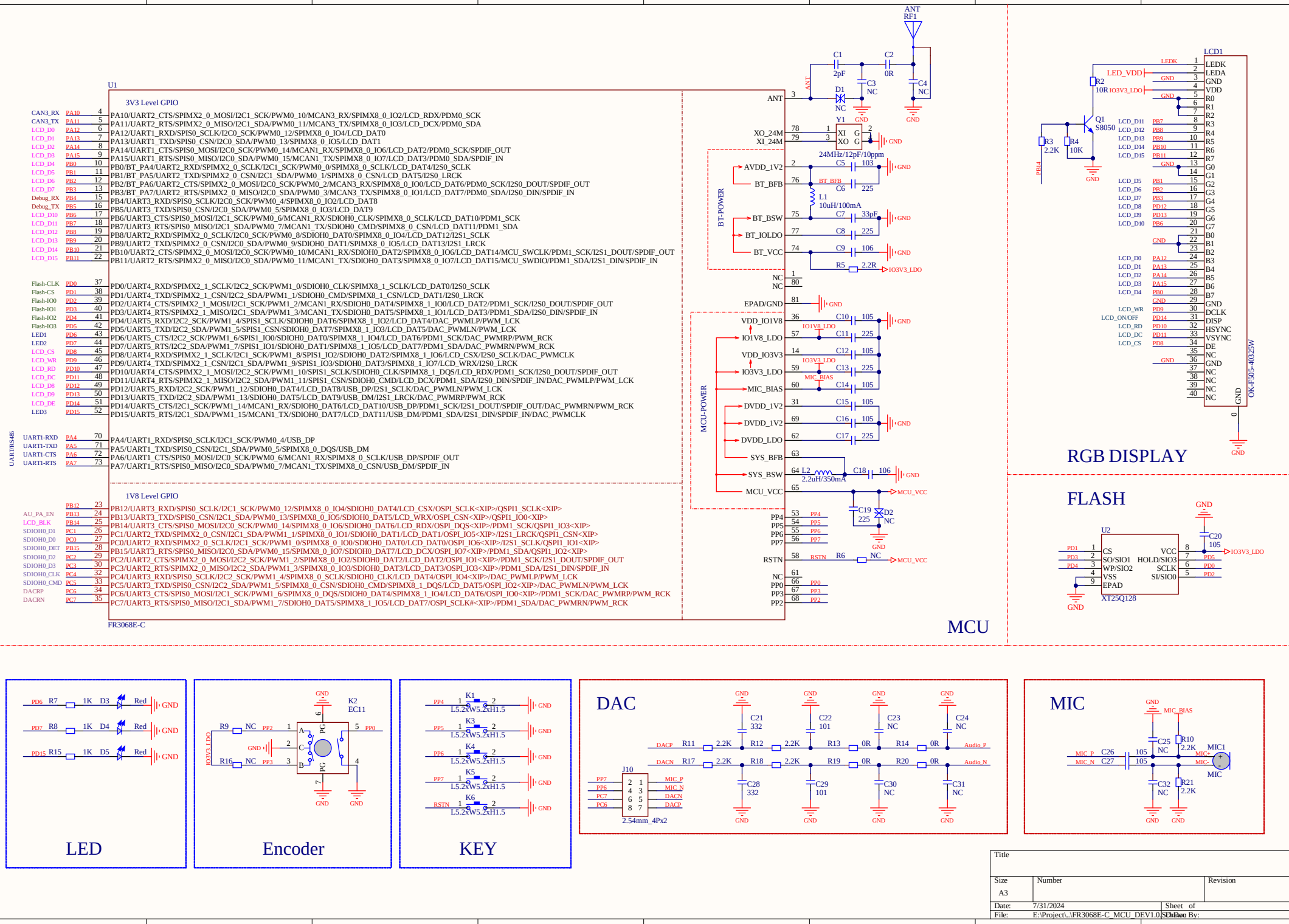
# 联系信息

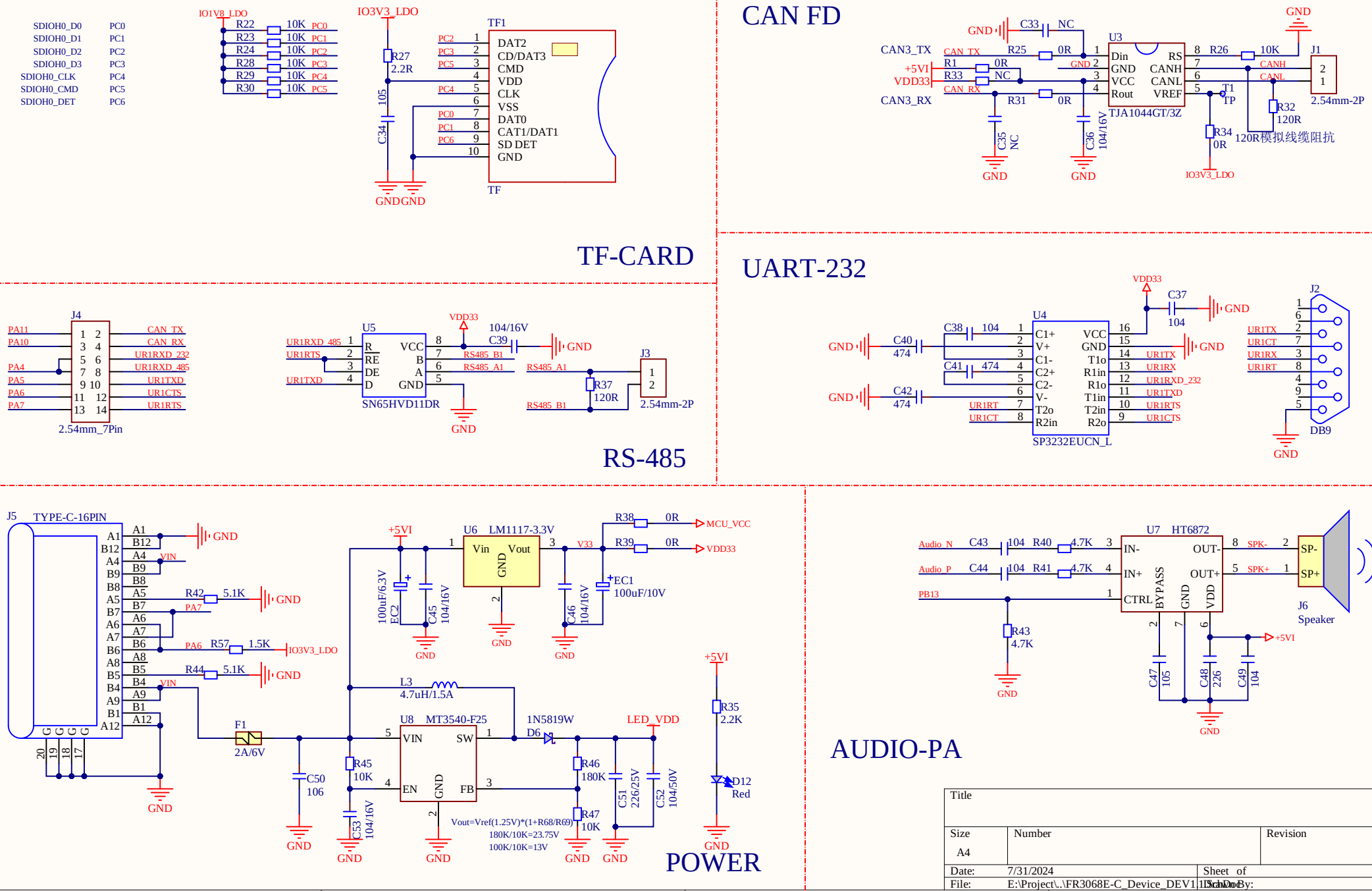
公司：上海富芮坤微电子有限公司
地址：中国(上海)自由贸易试验区碧波路 912 弄 8 号 501-A 室
电话：+86-21-5027-0080
网址: www.freqchip.com
销售邮箱: sales@freqchip.com
销售邮箱: docs@freqchip.com

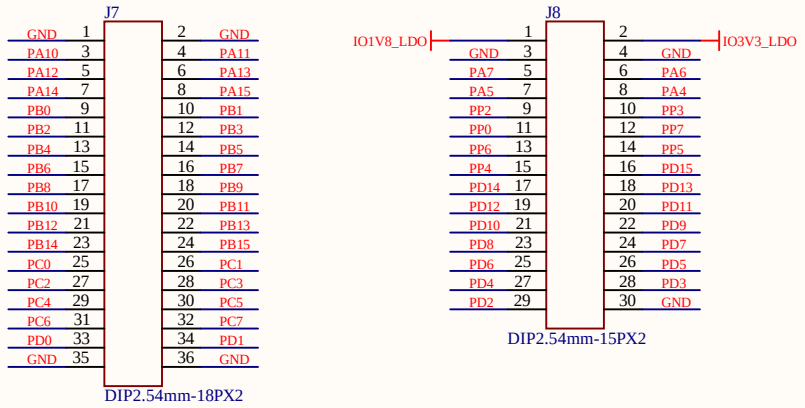
# 版本修订

| 版本号 | 发布日期 | 摘要 |
| --- | --- | --- |
| V0.1 | 2024.7.31 | 初版 |

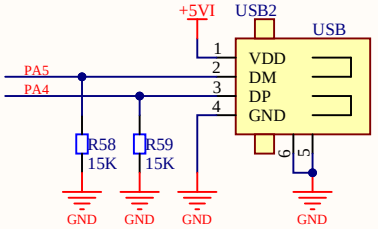
FreqChip Confidential


U1
3V3 Level GPIO
PA10/UART2_CTS/SPIMX2_0_MOSI/I2C1_SCK/PWM0_10/MCAN3_RX/SPIMX8_0_IO2/LCD_RDX/PDM0_SCK
PA11/UART2_RTS/SPIMX2_0_MISO/I2C1_SDA/PWM0_11/MCAN3_TX/SPIMX8_0_IO3/LCD_DCX/PDM0_SDA
PA12/UART1_RXD/SPIS0_SCLK/I2C0_SCK/PWM0_12/SPIMX8_0_IO4/LCD_DAT0
PA13/UART1_TXD/SPIS0_CSN/I2C0_SDA/PWM0_13/SPIMX8_0_IO5/LCD_DAT1
PA14/UART1_CTS/SPIS0_MOSI/I2C0_SCK/PWM0_14/MCAN1_RX/SPIMX8_0_IO6/LCD_DAT2/PDM0_SCK/SPDIF_OUT
PA15/UART1_RTS/SPIS0_MISO/I2C0_SDA/PWM0_15/MCAN1_TX/SPIMX8_0_IO7/LCD_DAT3/PDM0_SDA/SPDIF_IN
PB0/BT_PA4/UART2_RXD/SPIMX2_0_SCLK/I2C1_SCK/PWM0_0/SPIMX8_0_SCLK/LCD_DAT4/I2S0_SCLK
PB1/BT_PA5/UART2_TXD/SPIMX2_0_CSN/I2C1_SDA/PWM0_1/SPIMX8_0_CSN/LCD_DAT5/I2S0_LRCK
PB2/BT_PA6/UART2_CTS/SPIMX2_0_MOSI/I2C0_SCK/PWM0_2/MCAN3_RX/SPIMX8_0_IO0/LCD_DAT6/PDM0_SCK/I2S0_DOUT/SPDIF_OUT
PB3/BT_PA7/UART2_RTS/SPIMX2_0_MISO/I2C0_SDA/PWM0_3/MCAN3_TX/SPIMX8_0_IO1/LCD_DAT7/PDM0_SDA/I2S0_DIN/SPDIF_IN
PB4/UART3_RXD/SPIS0_SCLK/I2C0_SCK/PWM0_4/SPIMX8_0_IO2/LCD_DAT8
PB5/UART3_TXD/SPIS0_CSN/I2C0_SDA/PWM0_5/SPIMX8_0_IO3/LCD_DAT9
PB6/UART3_CTS/SPIS0_MOSI/I2C1_SCK/PWM0_6/MCAN1_RX/SDIOH0_CLK/SPIMX8_0_SCLK/LCD_DAT10/PDM1_SCK
PB7/UART3_RTS/SPIS0_MISO/I2C1_SDA/PWM0_7/MCAN1_TX/SDIOH0_CMD/SPIMX8_0_CSN/LCD_DAT11/PDM1_SDA
PB8/UART2_RXD/SPIMX2_0_SCLK/I2C0_SCK/PWM0_8/SDIOH0_DAT0/SPIMX8_0_IO4/LCD_DAT12/I2S1_SCLK
PB9/UART2_TXD/SPIMX2_0_CSN/I2C0_SDA/PWM0_9/SDIOH0_DAT1/SPIMX8_0_IO5/LCD_DAT13/I2S1_LRCK
PB10/UART2_CTS/SPIMX2_0_MOSI/I2C0_SCK/PWM0_10/MCAN1_RX/SDIOH0_DAT2/SPIMX8_0_IO6/LCD_DAT14/MCU_SWCLK/PDM1_SCK/I2S1_DOUT/SPDIF_OUT
PB11/UART2_RTS/SPIMX2_0_MISO/I2C0_SDA/PWM0_11/MCAN1_TX/SDIOH0_DAT3/SPIMX8_0_IO7/LCD_DAT15/MCU_SWDIO/PDM1_SDA/I2S1_DIN/SPDIF_IN
PD0/UART4_RXD/SPIMX2_1_SCLK/I2C2_SCK/PWM1_0/SDIOH0_CLK/SPIMX8_1_SCLK/LCD_DAT0/I2S0_SCLK
PD1/UART4_TXD/SPIMX2_1_CSN/I2C2_SDA/PWM1_1/SDIOH0_CMD/SPIMX8_1_CSN/LCD_DAT1/I2S0_LRCK
PD2/UART4_CTS/SPIMX2_1_MOSI/I2C1_SCK/PWM1_2/MCAN1_RX/SDIOH0_DAT4/SPIMX8_1_IO0/LCD_DAT2/PDM1_SCK/I2S0_DOUT/SPDIF_OUT
PD3/UART4_RTS/SPIMX2_1_MISO/I2C1_SDA/PWM1_3/MCAN1_TX/SDIOH0_DAT5/SPIMX8_1_IO1/LCD_DAT3/PDM1_SDA/I2S0_DIN/SPDIF_IN
PD4/UART5_RXD/I2C2_SCK/PWM1_4/SPIS1_SCLK/SDIOH0_DAT6/SPIMX8_1_IO2/LCD_DAT4/DAC_PWMLP/PWM_LCK
PD5/UART5_TXD/I2C2_SDA/PWM1_5/SPIS1_CSN/SDIOH0_DAT7/SPIMX8_1_IO3/LCD_DAT5/DAC_PWMLN/PWM_LCK
PD6/UART5_CTS/I2C2_SCK/PWM1_6/SPIS1_IO0/SDIOH0_DAT0/SPIMX8_1_IO4/LCD_DAT6/PDM1_SCK/DAC_PWMRP/PWM_RCK
PD7/UART5_RTS/I2C2_SDA/PWM1_7/SPIS1_IO1/SDIOH0_DAT1/SPIMX8_1_IO5/LCD_DAT7/PDM1_SDA/DAC_PWMRN/PWM_RCK
PD8/UART4_RXD/SPIMX2_1_SCLK/I2C1_SCK/PWM1_8/SPIS1_IO2/SDIOH0_DAT2/SPIMX8_1_IO6/LCD_CSX/I2S0_SCLK/DAC_PWMCLK
PD9/UART4_TXD/SPIMX2_1_CSN/I2C1_SDA/PWM1_9/SPIS1_IO3/SDIOH0_DAT3/SPIMX8_1_IO7/LCD_WRX/I2S0_LRCK
PD10/UART4_CTS/SPIMX2_1_MOSI/I2C2_SCK/PWM1_10/SPIS1_SCLK/SDIOH0_CLK/SPIMX8_1_DQS/LCD_RDX/PDM1_SCK/I2S0_DOUT/SPDIF_OUT
PD11/UART4_RTS/SPIMX2_1_MISO/I2C2_SDA/PWM1_11/SPIS1_CSN/SDIOH0_CMD/LCD_DCX/PDM1_SDA/I2S0_DIN/SPDIF_IN/DAC_PWMLP/PWM_LCK
PD12/UART5_RXD/I2C2_SCK/PWM1_12/SDIOH0_DAT4/LCD_DAT8/USB_DP/I2S1_SCLK/DAC_PWMLN/PWM_LCK
PD13/UART5_TXD/I2C2_SDA/PWM1_13/SDIOH0_DAT5/LCD_DAT9/USB_DM/I2S1_LRCK/DAC_PWMRP/PWM_RCK
PD14/UART5_CTS/I2C1_SCK/PWM1_14/MCAN1_RX/SDIOH0_DAT6/LCD_DAT10/USB_DP/PDM1_SCK/I2S1_DOUT/SPDIF_OUT/DAC_PWMRN/PWM_RCK
PD15/UART5_RTS/I2C1_SDA/PWM1_15/MCAN1_TX/SDIOH0_DAT7/LCD_DAT11/USB_DM/PDM1_SDA/I2S1_DIN/SPDIF_IN/DAC_PWMCLK
PA4/UART1_RXD/SPIS0_SCLK/I2C1_SCK/PWM0_4/USB_DP
PA5/UART1_TXD/SPIS0_CSN/I2C1_SDA/PWM0_5/SPIMX8_0_DQS/USB_DM
PA6/UART1_CTS/SPIS0_MOSI/I2C0_SCK/PWM0_6/MCAN1_RX/SPIMX8_0_SCLK/USB_DP/SPDIF_OUT
PA7/UART1_RTS/SPIS0_MISO/I2C0_SDA/PWM0_7/MCAN1_TX/SPIMX8_0_CSN/USB_DM/SPDIF_IN
1V8 Level GPIO
PB12/UART3_RXD/SPIS0_SCLK/I2C1_SCK/PWM0_12/SPIMX8_0_IO4/SDIOH0_DAT4/LCD_CSX/OSPI_SCLK<XIP>/QSPI1_SCLK<XIP>
PB13/UART3_TXD/SPIS0_CSN/I2C1_SDA/PWM0_13/SPIMX8_0_IO5/SDIOH0_DAT5/LCD_WRX/OSPI_CSN<XIP>/QSPI1_IO0<XIP>
PB14/UART3_CTS/SPIS0_MOSI/I2C0_SCK/PWM0_14/SPIMX8_0_IO6/SDIOH0_DAT6/LCD_RDX/OSPI_DQS<XIP>/PDM1_SCK/QSPI1_IO3<XIP>
PC1/UART2_TXD/SPIMX2_0_CSN/I2C1_SDA/PWM1_1/SPIMX8_0_IO1/SDIOH0_DAT1/LCD_DAT1/OSPI_IO5<XIP>/I2S1_LRCK/QSPI1_CSN<XIP>
PC0/UART2_RXD/SPIMX2_0_SCLK/I2C1_SCK/PWM1_0/SPIMX8_0_IO0/SDIOH0_DAT0/LCD_DAT0/OSPI_IO6<XIP>/I2S1_SCLK/QSPI1_IO1<XIP>
PB15/UART3_RTS/SPIS0_MISO/I2C0_SDA/PWM0_15/SPIMX8_0_IO7/SDIOH0_DAT7/LCD_DCX/OSPI_IO7<XIP>/PDM1_SDA/QSPI1_IO2<XIP>
PC2/UART2_CTS/SPIMX2_0_MOSI/I2C2_SCK/PWM1_2/SPIMX8_0_IO2/SDIOH0_DAT2/LCD_DAT2/OSPI_IO1<XIP>/PDM1_SCK/I2S1_DOUT/SPDIF_OUT
PC3/UART2_RTS/SPIMX2_0_MISO/I2C2_SDA/PWM1_3/SPIMX8_0_IO3/SDIOH0_DAT3/LCD_DAT3/OSPI_IO3<XIP>/PDM1_SDA/I2S1_DIN/SPDIF_IN
PC4/UART3_RXD/SPIS0_SCLK/I2C2_SCK/PWM1_4/SPIMX8_0_SCLK/SDIOH0_CLK/LCD_DAT4/OSPI_IO4<XIP>/DAC_PWMLP/PWM_LCK
PC5/UART3_TXD/SPIS0_CSN/I2C2_SDA/PWM1_5/SPIMX8_0_CSN/SDIOH0_CMD/SPIMX8_1_DQS/LCD_DAT5/OSPI_IO2<XIP>/DAC_PWMLN/PWM_LCK
PC6/UART3_CTS/SPIS0_MOSI/I2C1_SCK/PWM1_6/SPIMX8_0_DQS/SDIOH0_DAT4/SPIMX8_1_IO4/LCD_DAT6/OSPI_IO0<XIP>/PDM1_SCK/DAC_PWMRP/PWM_RCK
PC7/UART3_RTS/SPIS0_MISO/I2C1_SDA/PWM1_7/SDIOH0_DAT5/SPIMX8_1_IO5/LCD_DAT7/OSPI_SCLK#<XIP>/PDM1_SDA/DAC_PWMRN/PWM_RCK
FR3068E-C
MCU
ANT
RF1
C1 2pF
C2 0R
C3 NC
C4 NC
D1 NC
Y1
24MHz/12pF/10ppm
BT-POWER
L1 10uH/100mA
MCU-POWER
L2 2.2uH/350mA
D2 NC
R5 2.2R
R6 NC
IO3V3_LDO
IO1V8_LDO
MIC_BIAS
MCU_VCC
RGB DISPLAY
LCD1
OK-F505-40325W
R2 10R
R3 2.2K
R4 10K
Q1 S8050
LED_VDD
FLASH
U2
XT25Q128
C20 105
LED
Encoder
K2 EC11
KEY
DAC
J10
2.54mm_4Px2
C21 332
C22 101
C23 NC
C24 NC
C28 332
C29 101
C30 NC
C31 NC
MIC
C25 NC
C26 105
C27 105
C32 NC
R10 2.2K
R21 2.2K
MIC1
Title
Size
A3
Number
Revision
Date: 7/31/2024
Sheet of
File: E:\Project\..\FR3068E-C_MCU_DEV1.0_Sch.Doc
Drawn By:

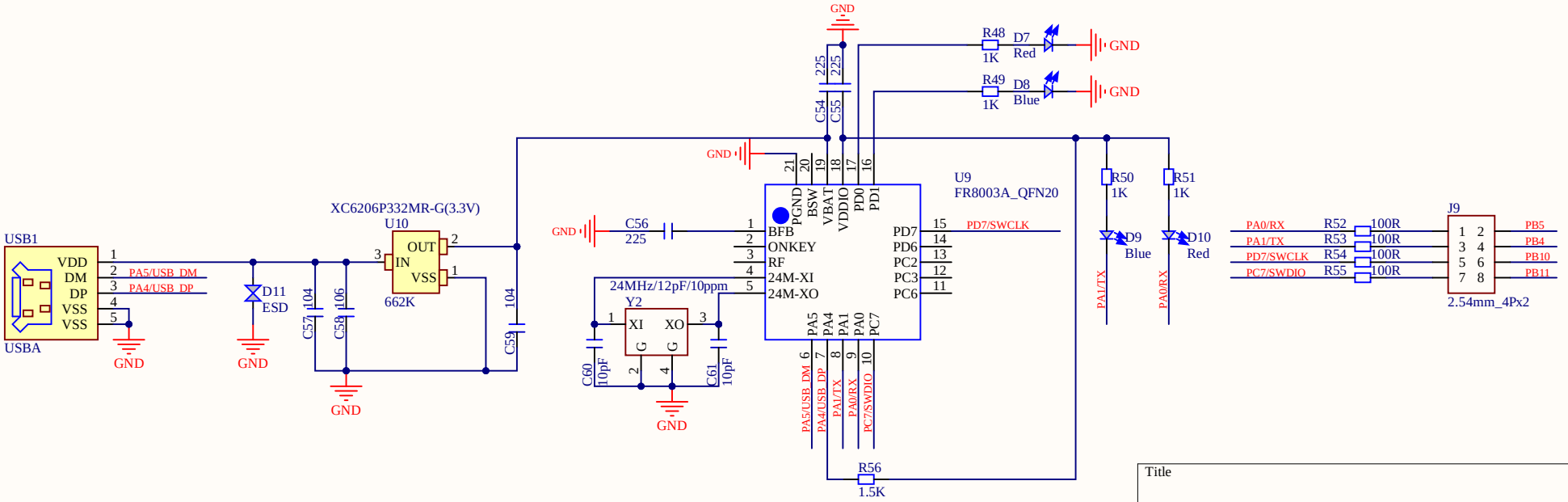

SDIOH0_D0 PC0
SDIOH0_D1 PC1
SDIOH0_D2 PC2
SDIOH0_D3 PC3
SDIOH0_CLK PC4
SDIOH0_CMD PC5
SDIOH0_DET PC6
IO1V8_LDO
R22 10K PC0
R23 10K PC1
R24 10K PC2
R28 10K PC3
R29 10K PC4
R30 10K PC5
IO3V3_LDO
R27 2.2R
C34 105
GNDGND
TF1
DAT2
CD/DAT3
CMD
VDD
CLK
VSS
DAT0
CAT1/DAT1
SD DET
GND
TF
TF-CARD
CAN FD
CAN3_TX
CAN3_RX
+5VI
VDD33
C33 NC
R25 0R
R1 0R
R33 NC
R31 0R
C35 NC
C36 104/16V
U3
Din
GND
VCC
Rout
RS
CANH
CANL
VREF
TJA1044GT/3Z
R26 10K
R32 120R
R34 0R
T1 TP
J1
2.54mm-2P
120R模拟线缆阻抗
GND
IO3V3_LDO
J4
PA11 CAN_TX
PA10 CAN_RX
PA4 UR1RXD_232
PA5 UR1RXD_485
PA6 UR1TXD
PA7 UR1CTS
UR1RTS
2.54mm_7Pin
U5
UR1RXD_485
UR1RTS
UR1TXD
R
RE
DE
D
VCC
B
A
GND
SN65HVD11DR
C39 104/16V
RS485_B1
RS485_A1
R37 120R
J3
2.54mm-2P
RS-485
UART-232
U4
C1+
V+
C1-
C2+
C2-
V-
T2o
R2in
VCC
GND
T1o
R1in
R1o
T1in
T2in
R2o
SP3232EUCN_L
C37 104
C38 104
C40 474
C41 474
C42 474
UR1TX
UR1RX
UR1RXD_232
UR1TXD
UR1RTS
UR1CTS
J2
DB9
J5 TYPE-C-16PIN
VIN
PA6
PA7
R42 5.1K
R44 5.1K
R57 1.5K
IO3V3_LDO
F1 2A/6V
U6 LM1117-3.3V
Vin
Vout
GND
V33
R38 0R
R39 0R
MCU_VCC
VDD33
EC1 100uF/10V
C45 104/16V
C46 104/16V
EC2 100uF/6.3V
L3 4.7uH/1.5A
U8 MT3540-F25
VIN
SW
EN
FB
GND
1N5819W
D6
LED_VDD
R45 10K
R46 180K
R47 10K
C50 106
C51 226/25V
C52 104/50V
C53 104/16V
Vout=Vref(1.25V)*(1+R68/R69)
180K/10K=23.75V
100K/10K=13V
R35 2.2K
D12 Red
POWER
Audio_N
Audio_P
PB13
C43 104
C44 104
R40 4.7K
R41 4.7K
R43 4.7K
U7 HT6872
IN-
IN+
CTRL
BYPASS
OUT-
OUT+
SP-
SP+
SPK-
SPK+
J6 Speaker
C47 105
C48 226
C49 104
AUDIO-PA
Title
Size A4
Number
Revision
Date: 7/31/2024
Sheet of
File: E:\Project\..\FR3068E-C_Device_DEV1.1\Sch.Doc
Drawn By:




MCU I/O

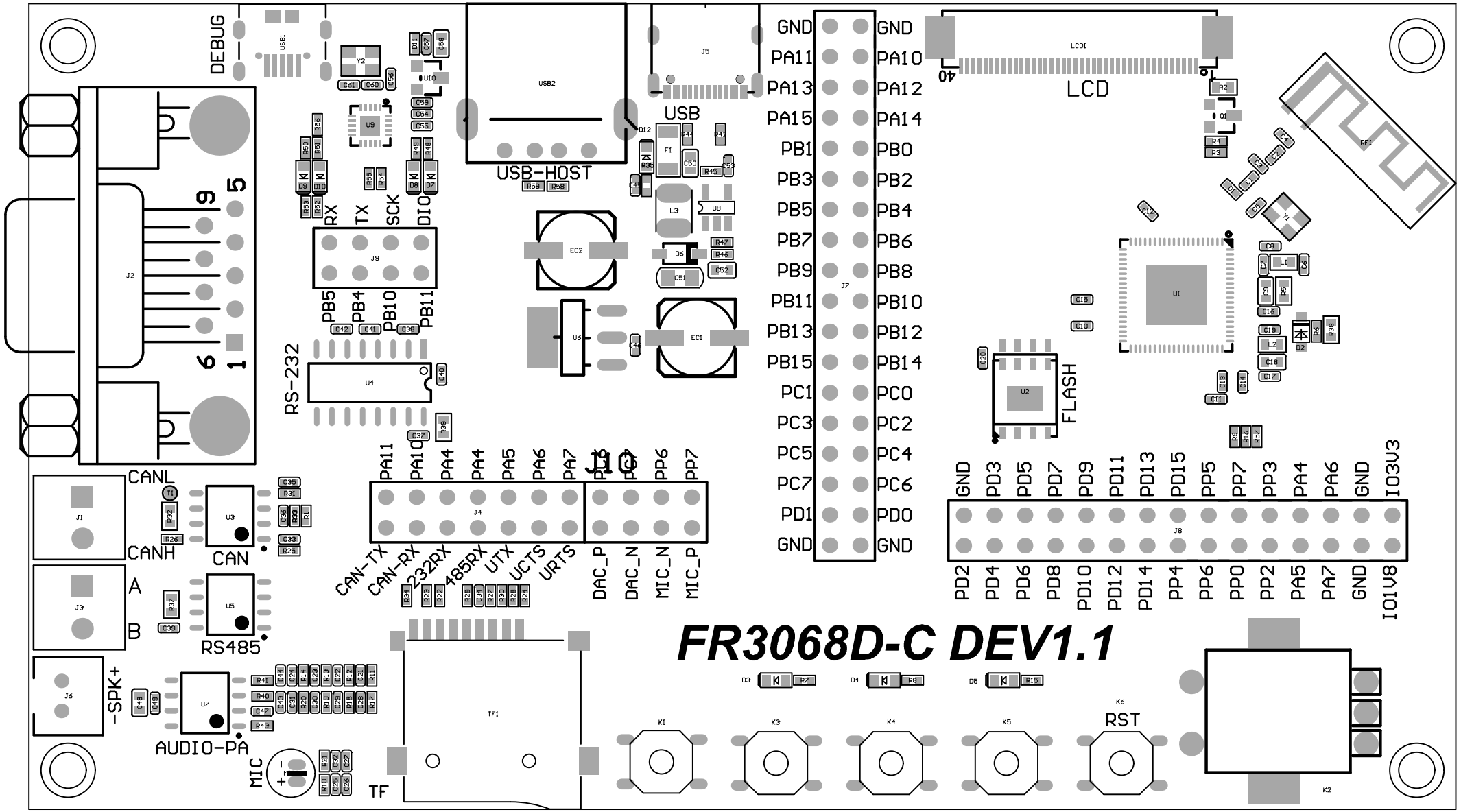



USB HOST




USB to TTL&DAP

| | | |
|---|---|---|
| Title | | |
| Size A4 | Number | Revision |
| Date: 7/31/2024 | Sheet of | |
| File: E:\Project\..\FR3068E-C_Debug_DEV1.0.SchDoc | Drawn By: | |




板层顺序：GTL>G1>G2>GBL